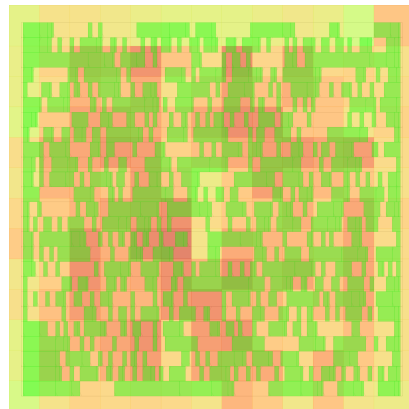
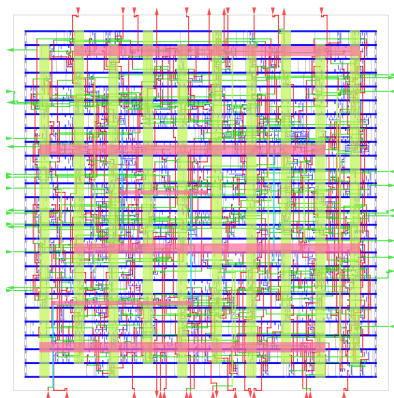


Filler cells and metal fill commands

```
# Close legal gaps in the standard-cell rows.
filler_placement {
    FILLCELL_X32 FILLCELL_X16 FILLCELL_X8
    FILLCELL_X4 FILLCELL_X2 FILLCELL_X1
}
check_placement -verbose

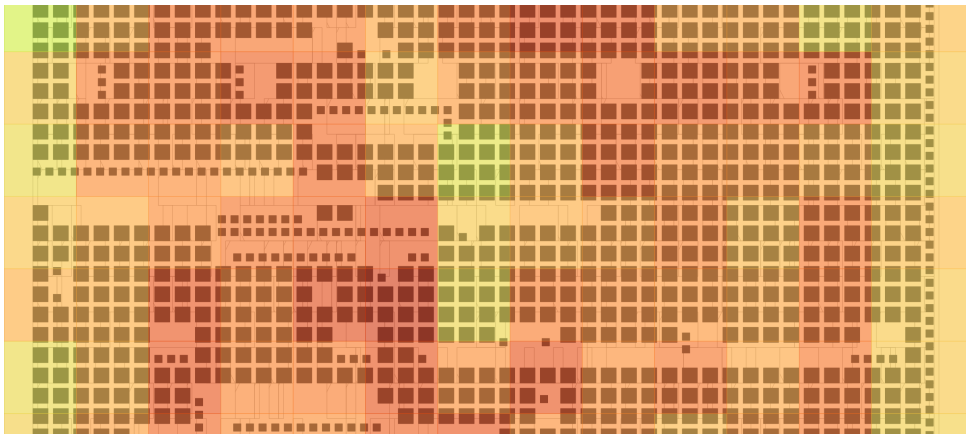
# Insert floating metal shapes using the PDK fill rules.
density_fill -rules fill.json
```

GCD after filler-cell placement



670 filler cells highlighted in green

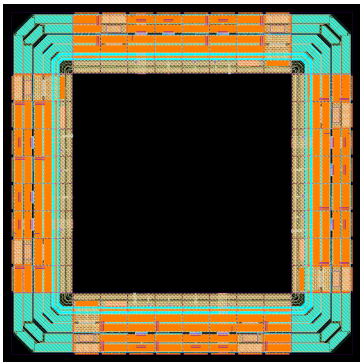
Metal fill on the GCD layout



1,895 added fill shapes on metal5.

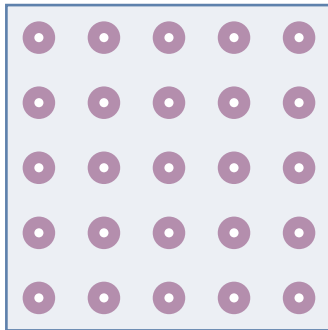
Top-level IO: pad rings and bump pads

Wire-bond pad ring



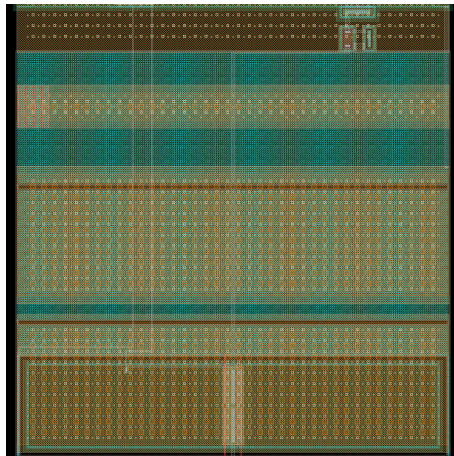
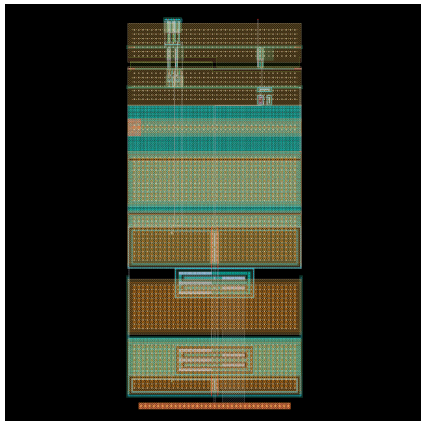
Bond wires leave the die at its edges.

Flip-chip bump pads



Bumps connect face-down across the die.

One IHP SG13G2 I/O pad



Final timing and power checks

Check the final routed design before exporting it.

```
# Report final setup and hold paths.
```

```
report_checks -path_delay min_max
```

```
# Check worst slack and total negative slack.
```

```
report_worst_slack -max
```

```
report_tns
```

```
# Check power and cell area (square micrometres).
```

```
report_power
```

```
report_design_area
```

Export the final design

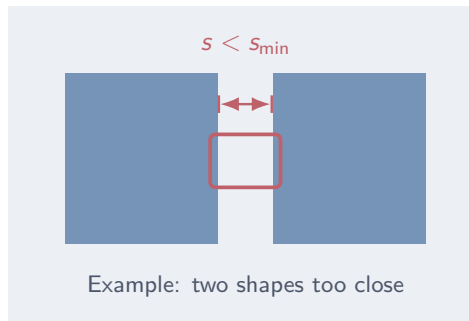
```
# OpenROAD database (.odb)
write_db final.odb
# Routed layout and placement (.def)
write_def final.def
# Cell and pin abstract (.lef)
write_abstract_lef final.lef
# Gate-level netlist (.v)
write_verilog -remove_cells {FILLCELL*} final.v
# Circuit netlist for LVS (.cdl); cells.cdl is the PDK master
write_cdl -masters {cells.cdl} final.cdl
# Timing constraints (.sdc)
write_sdc final.sdc
```

Convert DEF to GDS with def2stream.py

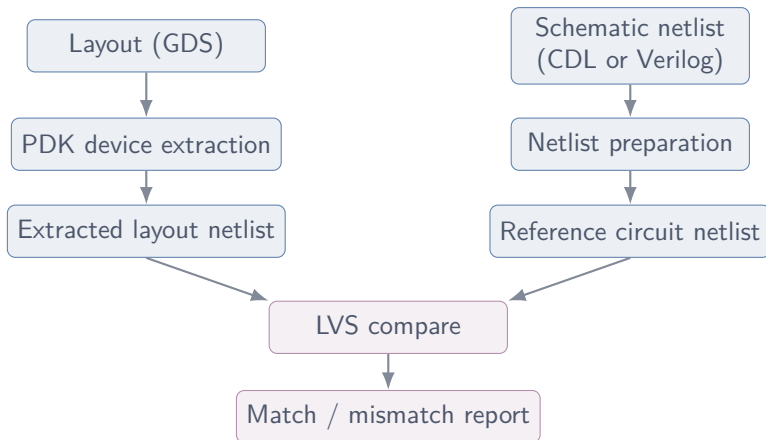
```
# Replace generic filenames and top with your design inputs.  
klayout -b -r scripts/def2stream.py \  
  -rd tech_file=platform.lyt \  
  -rd layer_map= \  
  -rd in_def=final.def \  
  -rd design_name=top \  
  -rd in_files="cells.gds macros.gds" \  
  -rd seal_file= \  
  -rd out_file=final.gds
```

Design rule checking (DRC)

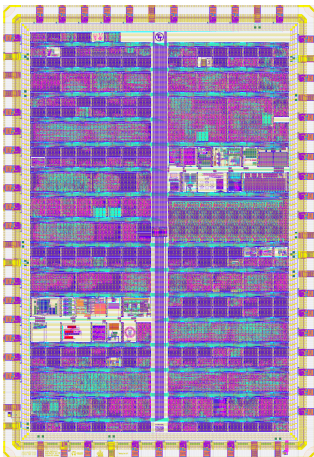
- Input** GDS + PDK DRC rule deck
- Check** Geometry against process rules
- Output** Markers and violation report



Layout versus schematic (LVS)



Tiny Tapeout: projects on one shared chip



TT SKY26c shared chip layout

~70 € per tile

One board and shipping: ~385 € total

app.tinytapeout.com/calculator